THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A triangle has sides 11 cm, 60 cm and 62 cm. Using the converse of the Baudhāyana–Pythagoras theorem, the triangle is: 1

(a) isosceles **(b)** right-angled **(c)** not right-angled **(d)** equilateral

2. For the expression $n^2 + n + 5$, the smallest natural number n for which the value is NOT prime is: 1

(a) 5 **(b)** 3 **(c)** 6 **(d)** 4

3. If a proposition and its converse are both true, we write: 1

(a) P if and only if Q **(b)** P or Q only
(c) P only **(d)** not P

4. Which of the following numbers has an odd number of factors? 1

(a) 34 **(b)** 30 **(c)** 36 **(d)** 37

5. The converse of the proposition "If a number is a multiple of 4, then it is even" is: 1

(a) If it is not true that it is even, then a number is a multiple of 4.
(b) If a number is a multiple of 4, then it is even.
(c) If a number is even, then it is a multiple of 4.
(d) If it is not true that a number is a multiple of 4, then it is not true that it is even.

6. The proposition "If a number is a multiple of 4, then it is even" is true. Its converse "If it is even, then a number is a multiple of 4" is: 1

(a) False **(b)** True only for some values
(c) True **(d)** It cannot be decided

7. Which of the following is a counterexample to the statement "The product of two irrational numbers is irrational"? 1

(a) $\sqrt{2} \times 1$ **(b)** $\sqrt{2} \times \sqrt{2}$
(c) $\sqrt{3} \times \sqrt{5}$ (irrational product $\sqrt{15}$) **(d)** $\sqrt{2} \times \sqrt{7}$ (irrational product $\sqrt{14}$)

8. The number of factors of 100 is: 1

(a) 11 **(b)** 9 **(c)** 8 **(d)** 10

9. A triangle has sides 12 cm, 35 cm and 37 cm. Using the converse of the Baudhāyana–Pythagoras theorem, the triangle is: 1

(a) not right-angled **(b)** right-angled **(c)** isosceles **(d)** equilateral

10. For the expression $n^2 + n + 41$, the smallest natural number n for which the value is NOT prime is: 1

(a) 39 **(b)** 40 **(c)** 41 **(d)** 42

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. How many factors does 81 have? What does the answer tell you about 81? 2

12. Write the converse of "If a number is prime and greater than 2, then it is odd". Is it true? 2

13. Give a counterexample to the statement "$n^2 - n + 11$ is prime for all natural numbers n". 2

14. How many factors does 90 have? What does the answer tell you about 90? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the converse of "If a number is prime and greater than 2, then it is odd". Is the converse true? Justify. 3

16. Show that the statement "Every quadrilateral with equal diagonals is a rectangle" is false by giving a counterexample. 3

17. Use the converse of the Baudhāyana–Pythagoras theorem to decide whether 18 cm, 24 cm, 31 cm are the sides of a right triangle. 3

OR

Write the converse of "If a number is a multiple of 8, then it is a multiple of 4". Is the converse true? Justify. 3

18. Show that the statement "If $x > y$, then $x^2 > y^2$" is false by giving a counterexample. 3

OR

Use the converse of the Baudhāyana–Pythagoras theorem to decide whether 28 cm, 45 cm, 54 cm are the sides of a right triangle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the sum of any 5 consecutive integers is divisible by 5. Verify for the numbers starting at 31. 5

OR

Prove that the square of an odd integer is odd and write the converse. Is the converse true? Test it with 13. 5

20. Prove that the sum of any 9 consecutive integers is divisible by 9. Verify for the numbers starting at 6. 5

OR

Prove that the square of an odd integer is odd and write the converse. Is the converse true? Test it with 21. 5

END OF QUESTION PAPER — Total: 40 Marks